

Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the 2N2222 NPN Bipolar Junction Transistor (BJT) can be modeled as a three-layer coupled soliton junction.

Rather than relying on quantum carrier wave-functions, the transistor's current gain (h_{FE}) and high-frequency switching limits are derived from the classical diffusion, recombination, and electric-field sweeping of mobile electron solitons [4.1] through a narrow, localized positive potential vacancy landscape [5.2].

Section 19: The Bipolar Junction Transistor (2N2222 Soliton Junction)

The 2N2222 is an NPN silicon planar switching transistor consisting of three distinct physical regions: the heavily doped Emitter, the narrow Base, and the moderately doped Collector.

[Emitter (N-type)] (High Density of Electron Solitons)	[Base (P-type)] (Narrow Region of Positive Vacancies)	[Collector (N-type)] (Moderately Doped with Electron Solitons)
Terminals: Emitter (E)	Base (B)	Collector (C)

19.1 Physical Architecture and Soliton Profiles In this system, the physical regions of the 2N2222 are modeled by their specific soliton distributions:

1. **Emitter (N-type):** A heavily doped semiconductor region with a high spatial density of mobile electron solitons (loops [4.1]).
2. **Base (P-type):** A very narrow, moderately doped region ($W_B \approx 0.5 \mu\text{m}$) characterized by localized positive core vacancies (holes [5.2]).
3. **Collector (N-type):** A moderately doped region containing mobile electron solitons, designed to absorb the carrier flow from the Base.

19.2 Base-Emitter (BE) Junction and Soliton Injection The base-emitter interface forms a depletion region of width $d_{BE} \sim \delta_{BE}$ [Section 3]. Applying a forward bias potential $V_{BE} > 0$ to the Base terminal shifts the local static potential barrier. Under the inhomogeneous field equation $D \cdot F_{\text{total}} = J_{\text{total}}$ [2.4], this lowers the electrostatic barrier at the junction.

Mobile electron solitons in the Emitter gain sufficient energy to overcome the remaining potential barrier and are injected across the BE junction into the Base. The injection current I_E is governed by the classical Boltzmann-like activation rate:

$$I_E \approx I_{ES} \left(e^{\frac{eV_{BE}}{k_B T}} - 1 \right)$$

where I_{ES} is the reverse saturation current of the BE junction.

19.3 Base Transport, Recombination, and Collector Sweep Once injected into the Base, the electron solitons act as minority carriers in a localized positive potential landscape.

19.3.1 Newtonian Diffusion Because the density of electron solitons is high at the BE boundary and zero at the collector-base (BC) boundary, a strong spatial concentration gradient is established. This gradient drives a classical, one-dimensional diffusion current of electron solitons across the Base width W_B .

19.3.2 Recombination (Base Current I_B) As the electron solitons diffuse through the Base, a small fraction of them collide with and fall into the localized positive core vacancies of the Base lattice, undergoing a local phase transition (recombination [5.2, 5.7]). This consumes base charge and constitutes the Base current:

$$I_B \approx \frac{Q_n}{\tau_p}$$

where: * Q_n is the total charge of the electron solitons currently in the Base.
 * τ_p is the physical recombination lifetime of the electron solitons in the base lattice.

19.3.3 Collector Sweep (Collector Current I_C) The collector-base (BC) junction is reverse-biased ($V_{CB} > 0$), establishing a strong static electric field E_{BC} pointing from Collector to Base. Because the physical width of the Base is extremely narrow compared to the diffusion length ($W_B \ll L_n$), more than 99% of the diffusing electron solitons reach the BC boundary before recombining.

Upon reaching this boundary, they are instantly swept across the BC depletion region into the Collector by the strong E_{BC} field, constituting the Collector current:

$$I_C \approx \frac{Q_n}{\tau_{\text{base}}}$$

where τ_{base} is the average transit time required for a soliton to diffuse across the Base.

19.4 Derivation of the Current Gain (h_{FE}) The static current gain h_{FE} (or β) of the 2N2222 is the ratio of the Collector current to the Base current:

$$h_{FE} = \beta = \frac{I_C}{I_B} = \frac{\tau_p}{\tau_{\text{base}}}$$

The average transit time τ_{base} for a soliton diffusing across the Base width W_B with diffusion coefficient D_n is:

$$\tau_{\text{base}} = \frac{W_B^2}{2D_n}$$

Substituting this into the gain equation:

$$h_{FE} = \frac{2D_n\tau_p}{W_B^2}$$

For the 2N2222, at typical operating currents ($I_C = 150$ mA, $V_{CE} = 10$ V), the narrow physical base width $W_B \approx 0.5$ μm ensures that the transit time is extremely short compared to the recombination lifetime ($\tau_{\text{base}} \ll \tau_p$), naturally yielding a high current gain:

$$h_{FE} \approx 100 \text{ to } 300$$

A small Base current I_B (modulating the charge density in the Base) controls a much larger Collector current I_C of electron solitons, realizing a current-controlled amplifier.

19.5 Transition Frequency (f_T) and Soliton Transit Limits The switching speed and high-frequency performance of the 2N2222 are limited by the physical transit times of the electron solitons through the device. The transition frequency f_T (the frequency at which the small-signal current gain extrapolates to unity) is:

$$f_T = \frac{1}{2\pi\tau_{EC}}$$

where τ_{EC} is the total emitter-to-collector delay time:

$$\tau_{EC} = \tau_E + \tau_{\text{base}} + \tau_C + \tau_D$$

- τ_E is the emitter charging time.
- τ_C is the collector depletion region charging time.
- τ_D is the collector drift delay.

For the 2N2222, the minimum transition frequency is:

$$f_T \approx 300 \text{ MHz}$$

This corresponds to a total physical soliton transit delay of:

$$\tau_{EC} = \frac{1}{2\pi \times (300 \times 10^6 \text{ s}^{-1})} \approx 0.53 \text{ nanoseconds}$$

This represents the physical, deterministic limit of the speed at which the electron loops [4.1] can be injected, transported, and swept through the 3D semiconductor crystal of the 2N2222, defining its high-speed switching and VHF amplification capabilities.